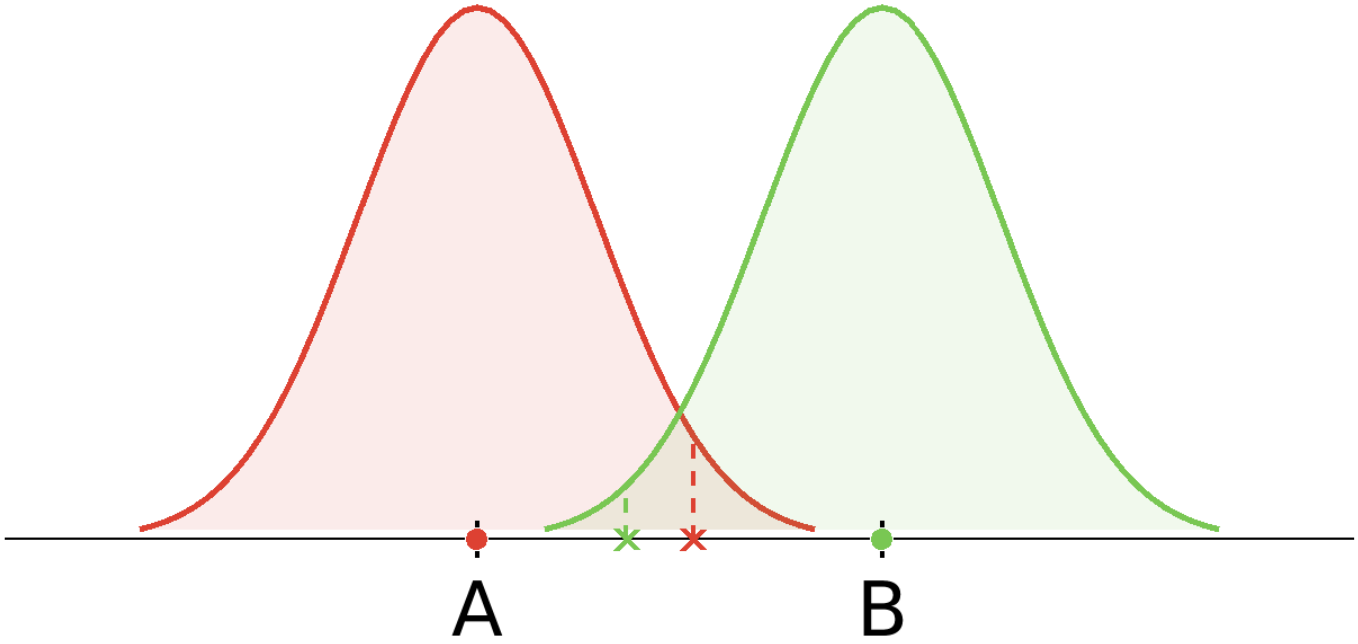
The Thurstone Scaling Model

Michael D. Lee

Law of Comparative Judgment (Thurstone, 1927)

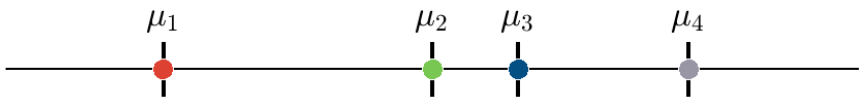
- The "law" of comparative judgment models how people judge which of two stimuli is "greater" in some perceptual dimension
  - brightness, duration, loudness, heaviness, ...
- True values lie on a latent dimension but have inherent "discriminal" variability that applies moment to moment
  - assumes people draw samples from each distribution and respond in terms of the greater sample



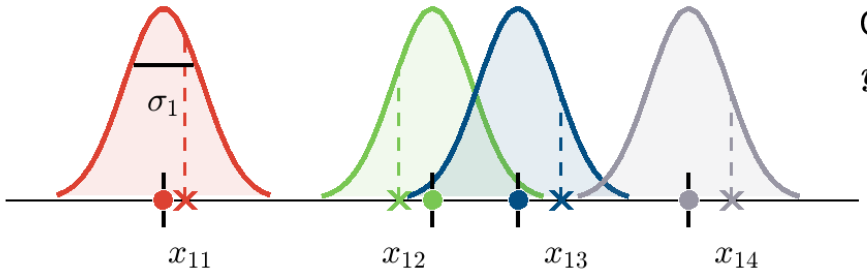
Thurstone Scaling

- The Thurstone scaling model applies this to ranking judgments
  - order the momentary samples from many stimuli
  - allow for differences in discriminal variability

Ground Truth

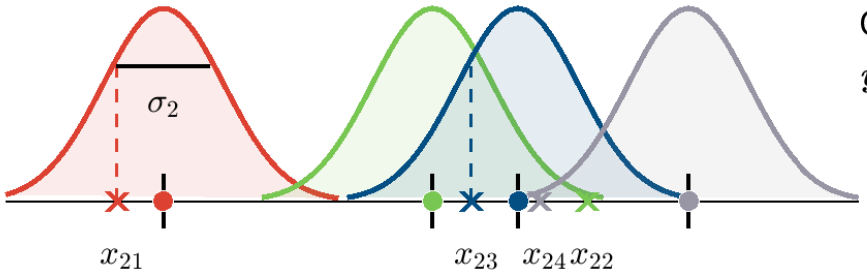


Individual One



Observed Ranking
$\boldsymbol{y}_1 = (1, 2, 3, 4)$

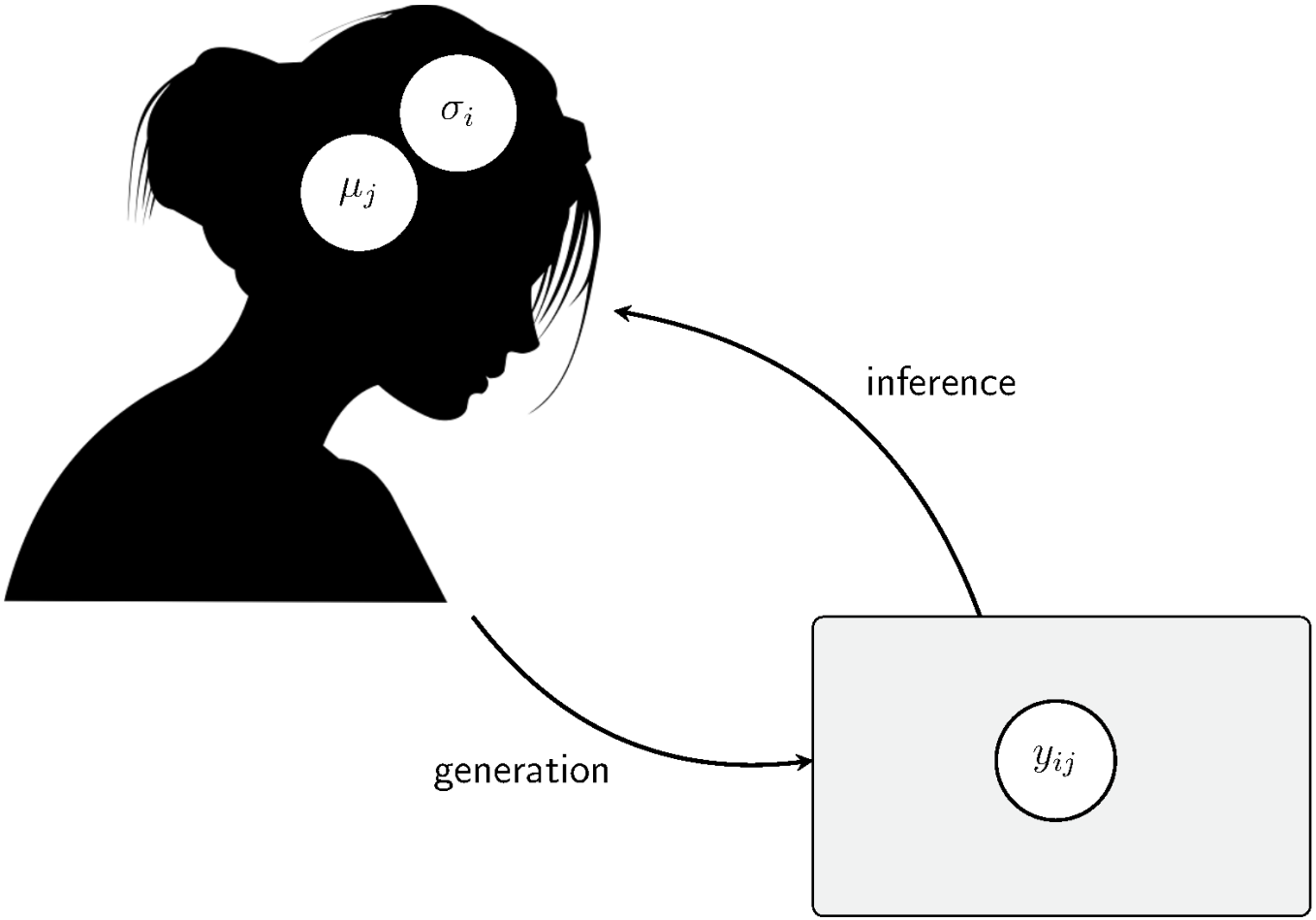
Individual Two



Observed Ranking
$\boldsymbol{y}_2 = (1, 3, 4, 2)$

Thurstone Scaling Model Structure

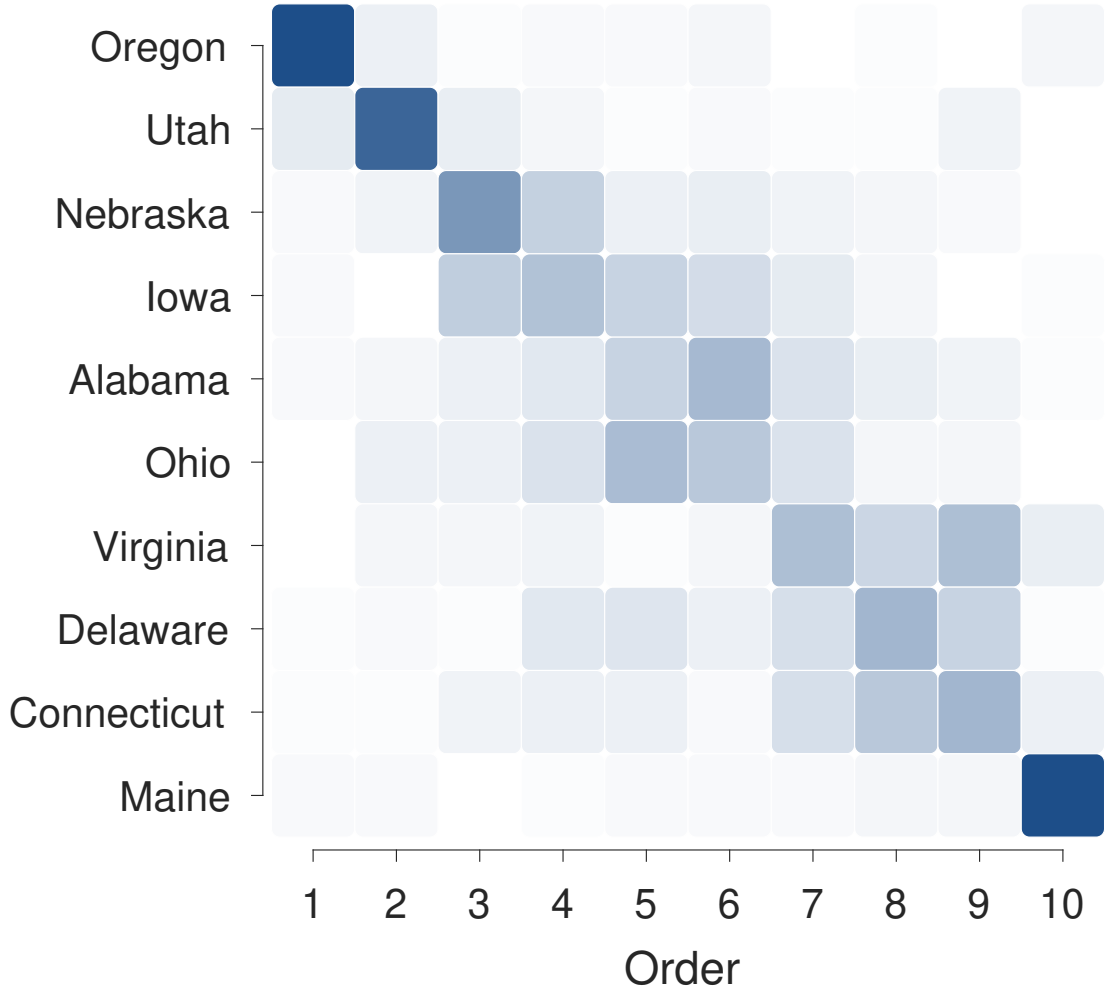
- The Thurstone Scaling model has the structure
  - observed ranking y_{ij} for stimulus j by individual i
  - representation μ_j parameter for each stimulus
  - discriminal variability (expertise) σ_i parameter for each individual



US States

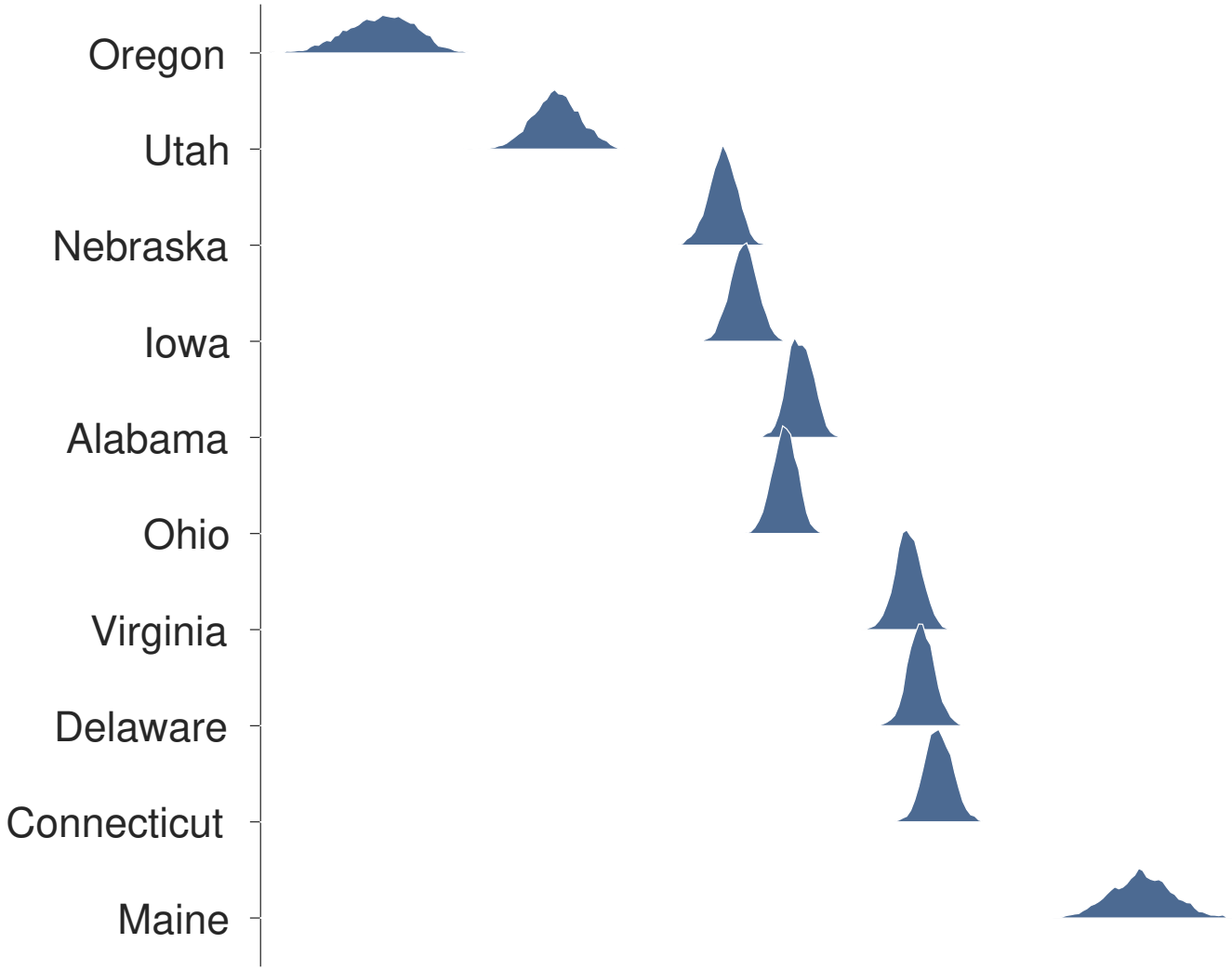
US States Data

- Lee, Steyvers, and Miller (2014) collected data from 78 participants ordering 10 US states geographically from the west to the east
  - the figure shows how often each state was ranked in each position



Inferred Representation

- The marginal posterior distributions of the μ_j representation locations are shown, in true order from top to bottom



Pairwise Probabilities

- The table reports the probability each stimulus (in a column) is greater than each other stimulus (in a row)

	Ore	Uta	Neb	Iow	Ohi	Ala	Vir	Del	Con	Mai
Oregon	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Utah	1.00	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nebraska	1.00	1.00	–	0.02	0.00	0.00	0.00	0.00	0.00	0.00
Iowa	1.00	1.00	0.98	–	0.00	0.00	0.00	0.00	0.00	0.00
Ohio	1.00	1.00	1.00	1.00	–	0.03	0.00	0.00	0.00	0.00
Alabama	1.00	1.00	1.00	1.00	0.97	–	0.00	0.00	0.00	0.00
Virginia	1.00	1.00	1.00	1.00	1.00	1.00	–	0.02	0.00	0.00
Delaware	1.00	1.00	1.00	1.00	1.00	1.00	0.98	–	0.01	0.00
Connecticut	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	–	0.00
Maine	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	–

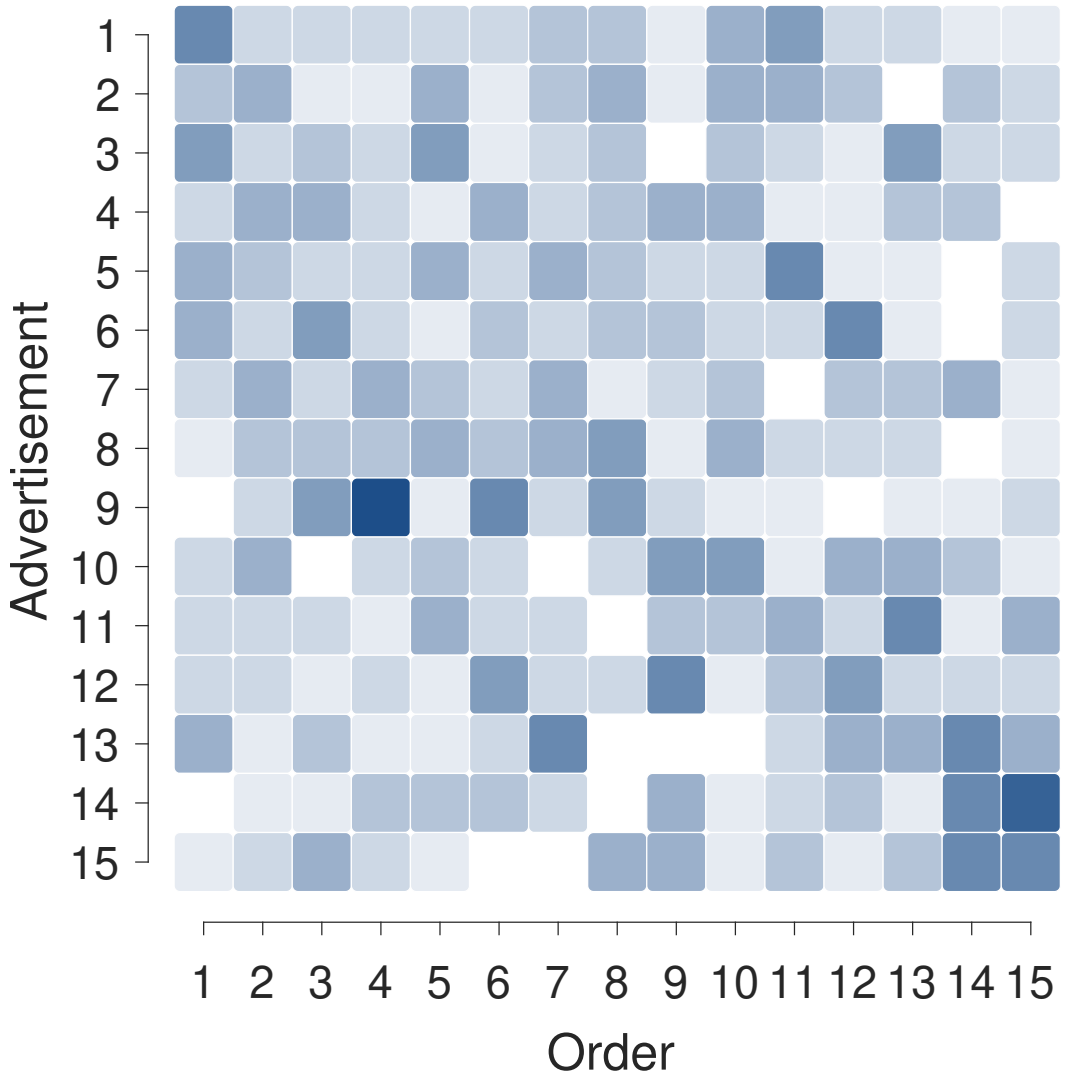
Inferences About Expertise

- The inferences about the expertise of each individual are provided by the posterior distributions for σ_i
- As an example, the rankings provided by two participants and their σ_i inferences are
  - participant 1: Oregon, Utah, Nebraska, Iowa, Delaware, Connecticut, Ohio, Alabama, Virginia, Maine
    - posterior mean is $\sigma_1 = 0.98$ with a 95% credible interval $(0.41, 2.00)$
  - participant 2: Connecticut, Alabama, Oregon, Delaware, Virginia, Iowa, Ohio, Nebraska, Utah, Maine
    - posterior mean is $\sigma_2 = 2.14$ with a 95% credible interval $(1.49, 2.63)$

Piano Advertisements

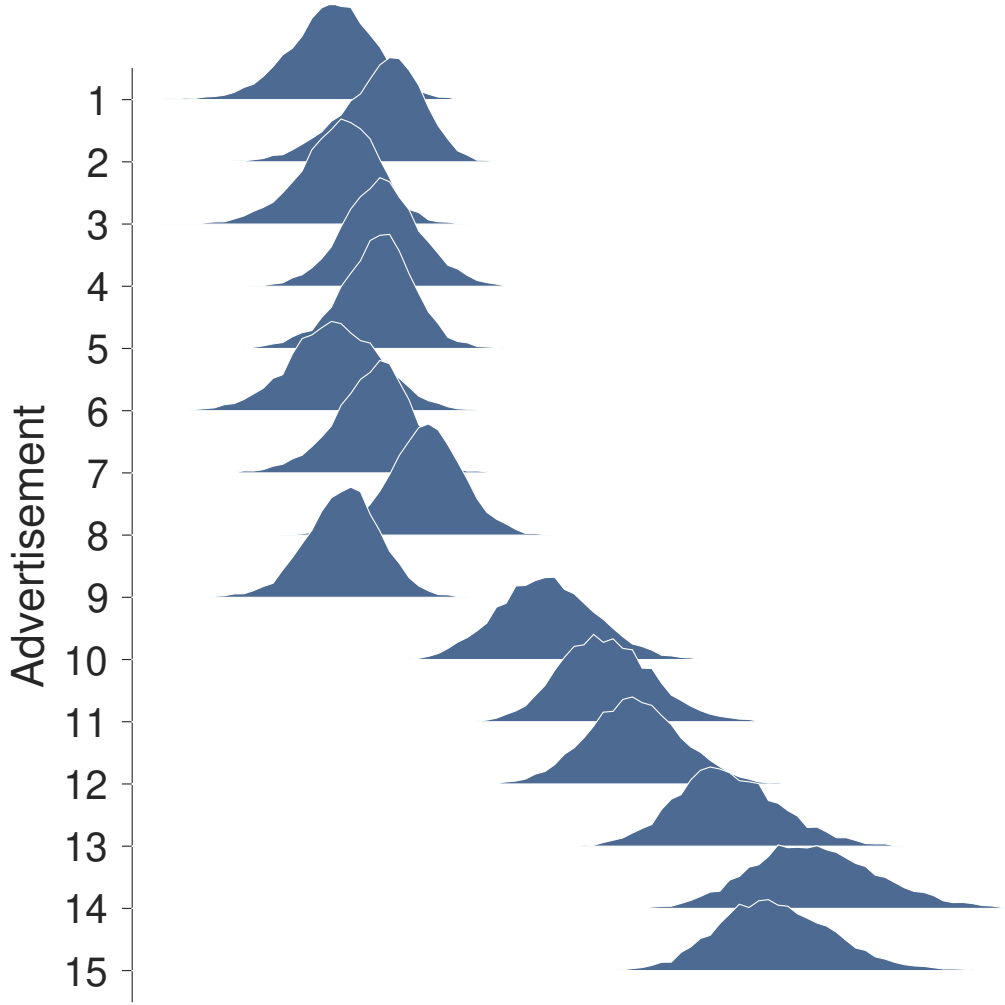
Piano Advertisements

- Knight (1921) collected rankings of the effectiveness of different advertisements for pianos



Inferred Representation

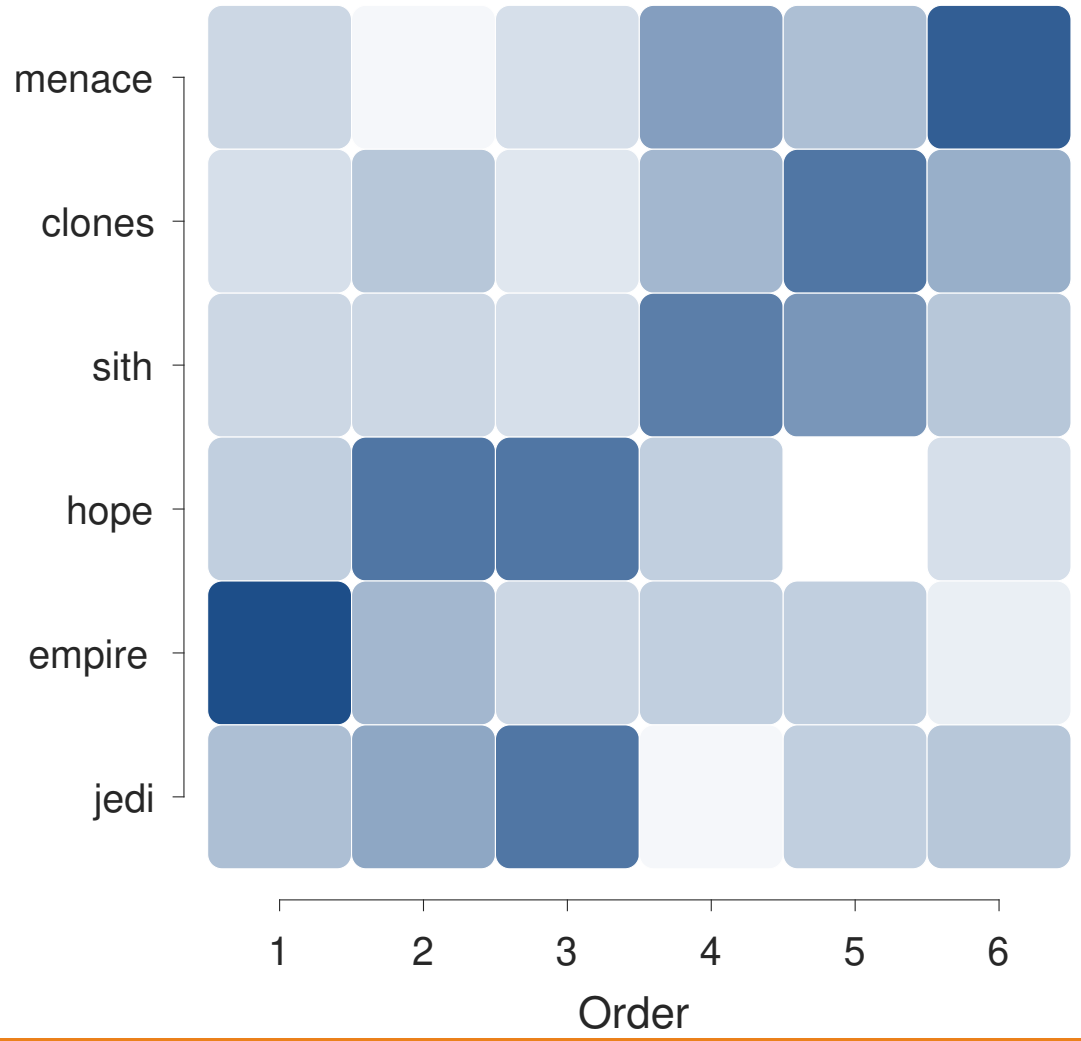
- The marginal posterior distributions of the μ_j representation locations are shown, in order of actual sales from top to bottom



Star Wars

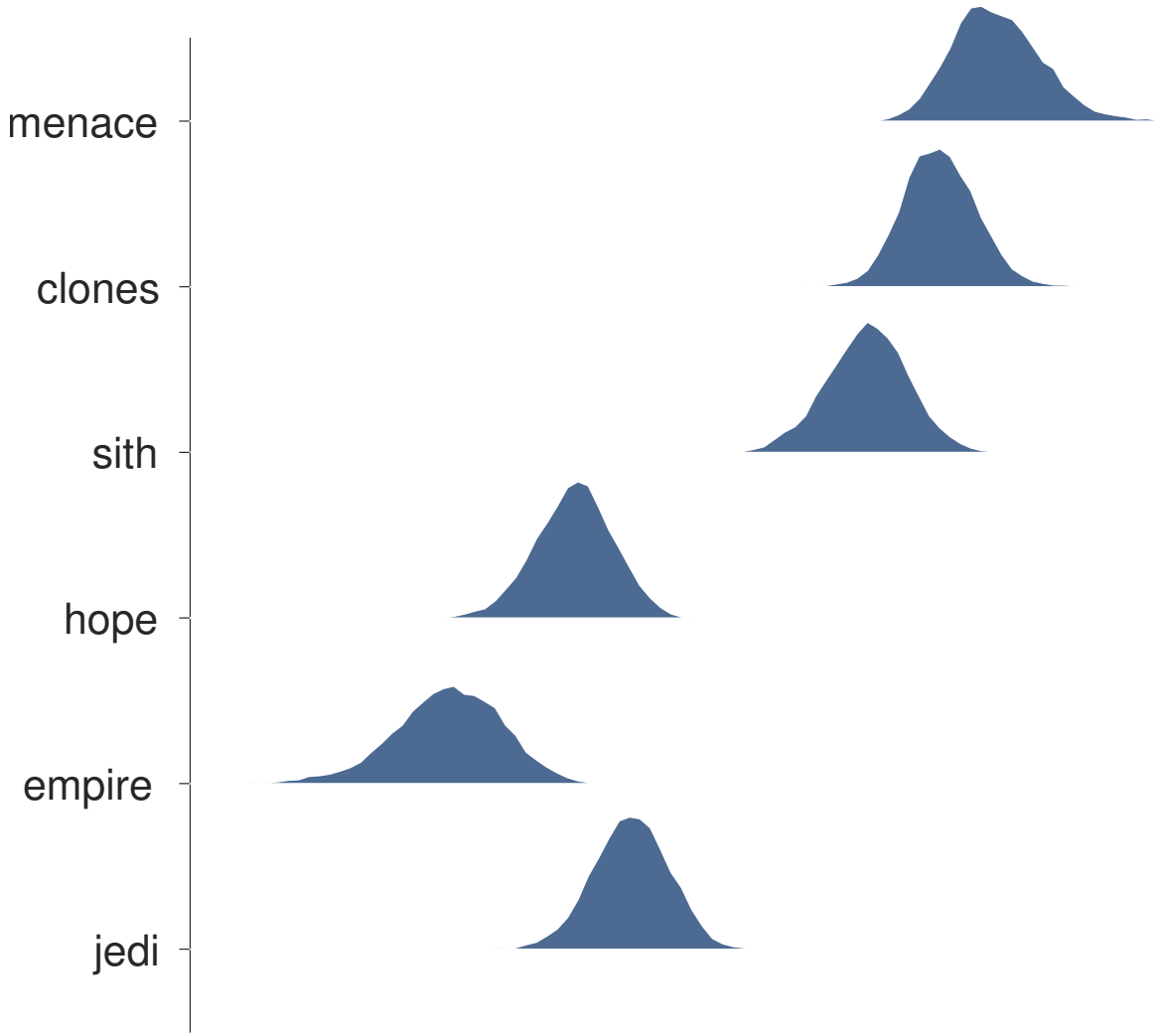
Star Wars Data

- Preference ratings from 50 Star Wars fans in a Kaggle data repository
  - ranking the first six movies of the Star Wars franchise



Inferred Representation

- The marginal posterior distributions of the μ_j representation locations in (Star Wars) chronological order from top to bottom



References i

Knight, H. C. (1921). *A comparison of the reliability of group and individual judgments* (Unpublished doctoral dissertation). Columbia University.

Lee, M. D., Steyvers, M., & Miller, B. J. (2014). A cognitive model for aggregating people's rankings. *PLoS ONE*, *9*, 1–9.

Thurstone, L. L. (1927). A law of comparative judgement. *Psychological Review*, *34*, 273–286.